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Conference and Expo



Using the Generic Conduit for Windows and Mac

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Question:

How do I write a conduit?

Answer:

Start with the Generic Conduit Framework.



Agenda Slide

- Conduit Fundamentals
- The Generic Conduit Framework
- Platform Independent Code/Features
- Platform Specific Code/Features
- New CDK for Mac OS X



Conduit Fundamentals

- One conduit per creator ID
- Windows and Mac conduits the use same SyncMgr API for communicating with the device
- Conduit registration is different on Mac and Windows
- There is no official support for Java conduits on the Mac



Conduit Fundamentals

- Several flavors of conduits
  - COM Sync Suite for developing conduits in Visual Basic (Windows only)
  - Conduit Development Kit, Java Edition for developing conduits in Java (Windows only)
  - Conduit Development Kit for developing conduits in C/C++ (Windows & Mac) including the Generic Conduit
- palmos.com/dev/tech/conduits/



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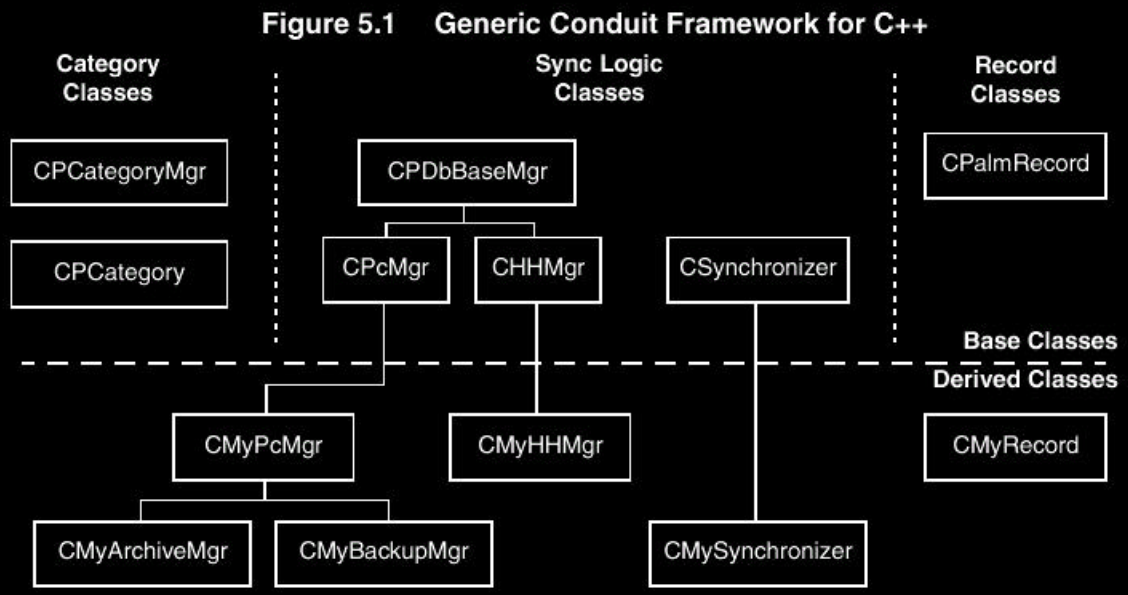
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The Generic Conduit Framework

- Cool Features
  - Will sync any database, regardless of format
  - Has sync logic built right in, just plug in your file format
  - X-platform code
  - Lots of utility classes to simplify development

The Generic Conduit Framework



From page 56 of the C/C++ Sync Suite Companion



The Generic Conduit Framework

```
ExportFunc long OpenConduit(PROGRESSFN pFn, CSyncProperties&
rProps)
{
    long retval = -1;
    if (pFn) {
        CSynchronizer* pGeneric;
        pGeneric = new CSynchronizer(rProps,
            GENERIC_FLAG_CATEGORY_SUPPORTED |
            GENERIC_FLAG_APPINFO_SUPPORTED);
         if (pGeneric){
           retval = pGeneric->Perform();
           delete pGeneric;
        }

    }
    return retval;
} // GenCond.cpp
```



The Generic Conduit Framework

```
SyncActionType fastHhActionArray[] = {
{HH_ATTR_DELETED | PC_ATTR_NO_REC,
 ACTION_DELETE_HH},

{HH_ATTR_DELETED | PC_ATTR_NONE,
 ACTION_DELETE_HH | ACTION_DELETE_PC},

{HH_ATTR_DELETED | PC_ATTR_DELETED,
 ACTION_DELETE_HH | ACTION_DELETE_PC},

{HH_ATTR_DELETED    | PC_ATTR_ARCHIVED,
 ACTION_DELETE_HH | ACTION_ARCHIVE_PC},
… // SyncActn.cpp
```



The Generic Conduit Framework

- Customize File Format in CPcMgr
  - Overwrite RetrieveDB, ConvertPCtoGeneric, StoreDB, etc
  - Use X-platform calls like WriteSerializedString
- Subclass Csynchronizer to use custom CPcMgr
  - Overwrite CreatePCManager
- May want to subclass CHHMgr if you do interesting things in the AppInfoBlock
  - IE Address Book



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Platform Independent Code/Features

- Palm Foundation Classes (PFC)
  - X-platform utility classes (CPString, CPFile, etc)
  - Useful Palm classes (CPalmRec, CPCategory, etc)
  - Don’t have to use Generic Conduit to use PFC



Platform Independent Code/Features

- Generic Conduit does the “right thing” on both Mac/Windows
  - Different entry points for Mac/Windows
  - Different UI for Mac/Windows
  - Uses #ifdef, #else, #endif to encapsulate platform specific code for you

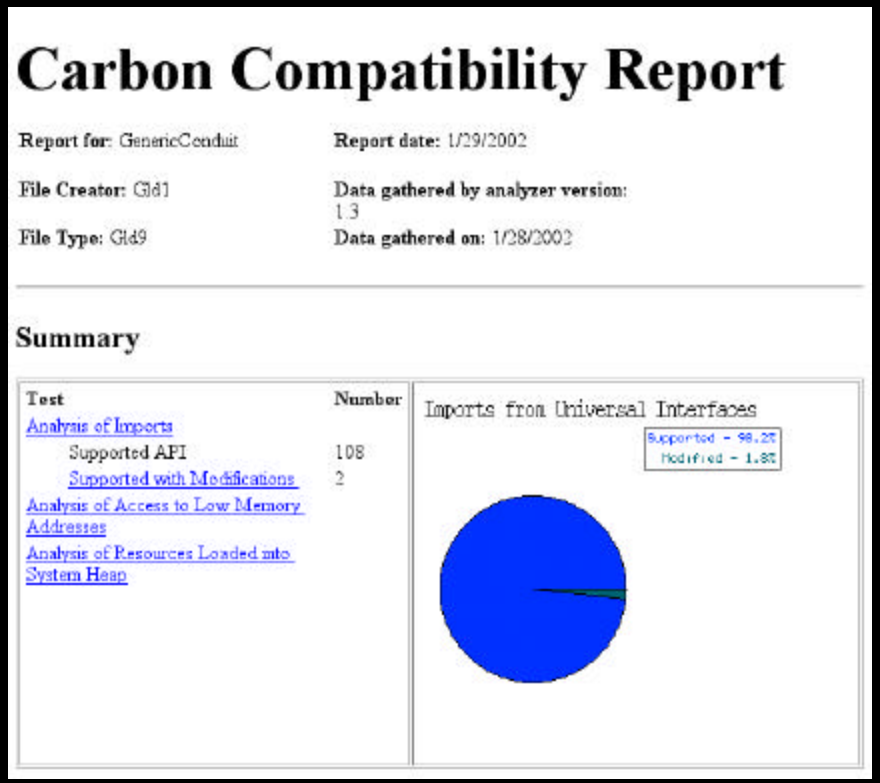


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Platform Specific Code/Features

- Lots of new new features for Mac OS X
  - Carbonized the Generic Conduit



Carbon Compatibility Report

Report for: GenericConduit
Report date: 1/29/2002
File Creator: Gld1
Data gathered by analyzer version: 1.3
File Type: Gld9
Data gathered on: 1/28/2002

Summary

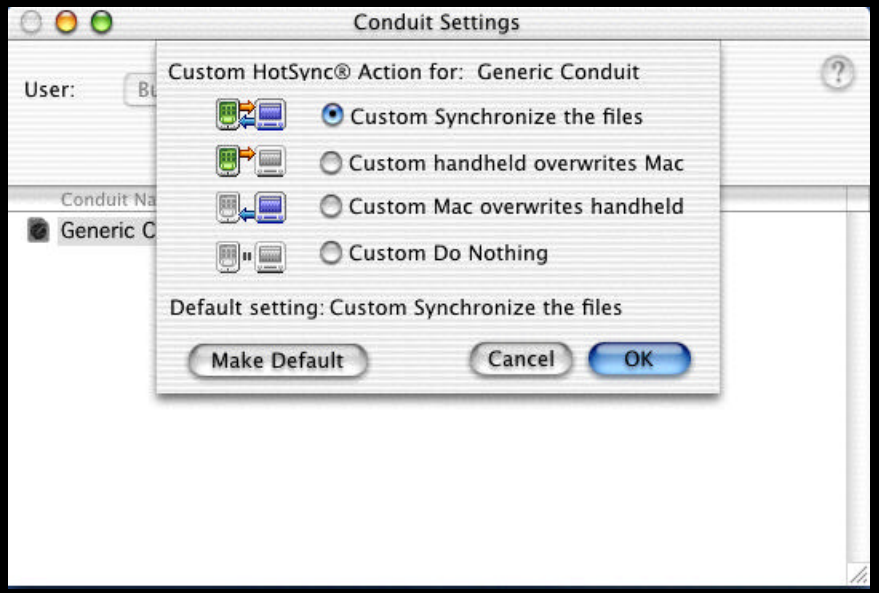
Test	Number
Analysis of Imports	
Supported API	108
Supported with Modifications	2
Analysis of Access to Low Memory Addresses	
Analysis of Resources Loaded into System Heap	

Imports from Universal Interfaces

Supported - 98.2%
Modified - 1.8%

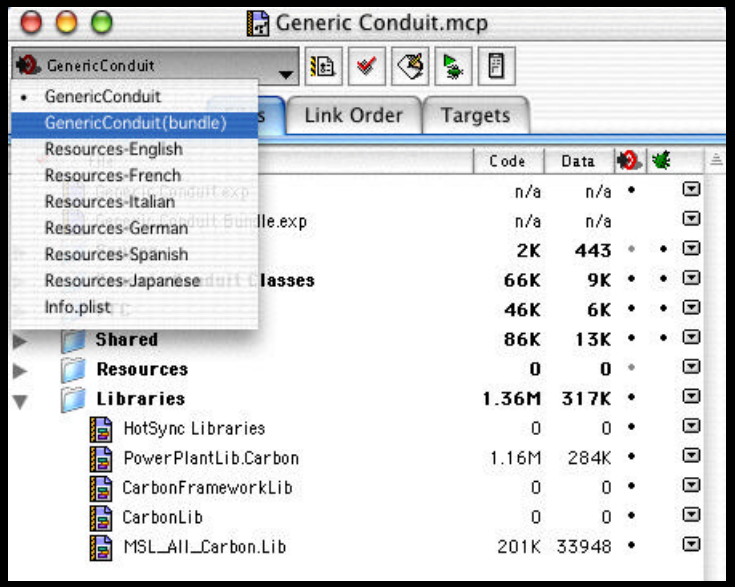
Platform Specific Code/Features

- Lots of new new features for Mac OS X
  - Aquified the Generic Conduit custom UI



Platform Specific Code/Features

- Lots of new new features for Mac OS X
  - New Generic Conduit bundle target





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New CDK for Mac OS X

- DR1 is available as of PalmSource
- Download from Palm OS Developer Seeding Area in the Resource Pavilion
  - www.palmos.com/dev/programs/pdp/login.html
- Please download and give feedback while we can still incorporate changes



Questions? Comments? Snide Remarks?

- www.palmos.com/dev/programs/pdp/login.html



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